Princess Jaena Marie O. De La Peña

Computer Science, University of the Philippines · Operations, automation & immersive tech

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Profile

Computer Science student at the University of the Philippines (class of 2027) working where operations, software, and 3D meet. At Belli I automate air-cargo and finance workflows; at ISCOLab (UPOU) I build VR experiences and optimize 3D assets as a Junior Research Prefect. I have shipped web apps, real-time 3D worlds in the browser, and game-jam games, and I lead partnerships for the UP Computer Science Guild. I am currently building The Garden of Almira, a walkable 3D garden of Cebu's herbal knowledge and oral tradition that I am developing into a research paper.

Experience

Operations Associate · Belli — 2026 – Present

- Streamline and automate air-cargo operations workflows, applying technical problem-solving to systemize supply-chain logistics.
- Built the Grab Receipts Matcher, a browser tool that reconciles Grab for Business receipts with Brex card charges for month-end accounting.
- Oversee back-end administrative processes, workflow automation, and digital strategy execution to improve operational efficiency.
- Completed onboarding training in Singapore.

Operations Manager & Automation Specialist · VERGE Inc, Watertown, South Dakota (Remote) — Jul 2024 – Jan 2026

- Oversaw back-end operations, workflow automation, and human resources administration.
- Drove digital growth and lead generation through SEO and digital marketing strategy.
- Built automated internal tools using Visual Basic and website-builder platforms.

Research & programs

Junior Research Prefect · ISCOLab (Immersive Science Communication Open Laboratory), University of the Philippines Open University — Dec 2025 – Present

- Developing BinItBetter (ongoing), a gamified VR sustainability application set in the UP OUverse that teaches proper waste segregation.
- Create spatial-computing and immersive-technology prototypes aligned with the UN Sustainable Development Goals (SDGs).
- Optimize 3D assets (meshes, materials, and LOD models) in Blender and Unity for spatial-computing environments.

Research Intern · ISCOLab, UPOU, Los Baños, Laguna — Jun – Jul 2026

- Completed a 240-hour onsite on-the-job training building VR and AR projects for science communication, sustainability, and immersive learning.
- Built *How a Nation Was Made*, a 3D web game on the origins of the Philippines where players sail a balangay between islands and make choices that shape the archipelago.
- Built "The Human Engine," an MSVR prototype for the MSVR headset: an immersive anatomy tour of the heart, lungs, and blood vessels with quiz checkpoints.
- Co-developed BinItBetter and presented our work at "Spatial Shift," the lab's first public event (July 29, 2026).

Participant & App Designer · UP – University of Tokyo Virtual Exchange Program (CALL) — Apr – Jul 2026

- Designed Computer-Assisted Language Learning (CALL) app proposals with cross-cultural teams from UPOU, UP Cebu, and the University of Tokyo, applying pedagogical theory and user-centered design; pitched at ISCOLab in July 2026.

Education

B.S. Computer Science · University of the Philippines Cebu, Cebu City — 2023 – Expected 2027

- College Scholar
- DOST-SEI S&T Undergraduate Scholarship (2023), Department of Science and Technology

Selected projects

The Garden of Almira · 3D garden of Visayan herbal knowledge — 2026 – Present
Ongoing, research paper in progress · Creator & 3D Developer · Babylon.js, JavaScript, WebGL, Web Audio · garden-of-almira.vercel.app

- Building a walkable 3D folk garden of Sugbo (Cebu), inspired by the title *Garden of Almira* by Erlinda Alburo: 18 native flowers, each with its Cebuano and scientific names and its place in Visayan herbal practice and oral tradition.
- Keeps it real-time in the browser with GPU vertex wind, three levels of detail per plant, distance culling, chunked grass, and a live-generated soundscape with no audio files.
- Writing a research paper based on the project (in progress).

Sulat ni Jae · a journal where every page grows its own 3D world — 2026
Live · Creator & Developer · Next.js, three.js, NLP from scratch, Supabase · sulat-ni-jae.vercel.app

GreenPoint · GIS-based framework for greening interventions — 2025 – 2026
Research & Innovation · Co-Creator · fp-green-point.vercel.app · Champion, Software Engineering Project Expo 2026 · Finalist, Aboitiz Innovate Cebu 2026, PJDSC 2025, PhilSA 2025

- Co-developing a GIS application that analyzes Urban Heat Island (UHI) effects and gives AI-driven greening recommendations to mitigate urban climate impacts.
- Led the initial design-thinking phase: urban environmental research, spatial-data ideation, and the technical paper pitched across data science and innovation competitions.

FREC Monitoring System · CertTrack, certification workflow tracking — 2026
ISCOLab (UPOU) side project · Developer · React 19, Vite, Tailwind CSS, Express, Prisma, PostgreSQL

- Building a role-based system where students submit theses, research papers, and projects for certification and faculty roles (adviser, FREC, program chair, dean, reviewer) process them.
- Routes each submission through one of three adviser-selected approval chains, from certificate generation to dean endorsement and final FICS FREC confirmation.

Grab Receipts Matcher · receipt-to-card reconciliation tool — 2026
Automation, built at Belli · Creator · JavaScript, jsPDF, SheetJS

- Turns Grab for Business exports into PDF receipts and matches each trip to its Brex charge by booking ID, amount, and time, including split charges and refunds.
- Outputs a sorted Excel masterlist with QuickBooks and Brex links and ready-to-paste memos, replacing manual month-end matching.

Oki Ventures · home services booking & management system — Ongoing
Startup Web Platform · Co-Creator & Full-Stack Developer

- Developing a web platform for booking and managing home services, connecting clients with service workers.

IBiz · business planning platform for women entrepreneurs — Jul 2025
Hackathon, IBPAP 2025 · Co-Creator · React, Vite, Tailwind CSS, JavaScript

- Built a web app that helps women entrepreneurs refine ideas, generate feasibility reports, and write business plans, with startup guides and a community space.

PWDIndependence · emergency app for persons with disabilities — Mar 2025
Hackathon, Komsai Week 2025 · Co-Creator · Python, Flet, SQLAlchemy, Pandas, Scikit-learn (DBSCAN)

- Built a distress-signal app with one-button emergency alerts, real-time location tracking, emergency-contact notifications, and medical-info access.

UNTREAD · 2D game — 2026
Game Jam, UPCSG GameJam 2026 · Game Developer · Godot

- Co-created a 2D game around the theme “The Path Less Taken” within the game-jam timeframe.

Hamuketsu · comedy-driven 2D game — Jun 2025
Game Project · Co-Creator · HTML, JavaScript, CSS

- Developed a pixel-art browser game with themed minigames, using finite state machines for branching choices, puzzles, and escalating difficulty.

FabLearn XR · semiconductor learning prototype
3D Web, Educational · 3D Web Developer · Three.js, HTML5, JavaScript

- Built a lightweight interactive 3D demo with step-by-step learning modules and quizzes.

COOKBOOk · recipe app with a pantry system — Nov – Dec 2024

Academic Project, Data Structures · Co-Creator · Python, Flet, NLTK

- Suggests recipes from available ingredients using tries, hash tables, scapegoat trees, and linked lists, with NLTK lemmatization of ingredients.

Split & Kanban Board · software engineering projects

Academic Projects, Software Engineering · Web Engineering

- Split: collaborative expense manager with interactive tracking cards and offline synchronization.
- Kanban Board: task manager with drag-and-drop workflow tracking and a responsive UI.

Technical skills

Web & full-stack	React, React Native, Next.js, Vite, Flutter, Flet, Tailwind CSS, HTML5, CSS3
Backend & data	Supabase, Prisma, PostgreSQL, SQLAlchemy, Pandas, Scikit-learn
Languages	TypeScript, JavaScript, Python, C, C++, C#, LaTeX
3D & games	Blender (LOD modeling, mesh optimization, materials, animation, game assets), Unity, Godot, Three.js, Babylon.js
Design	Figma (UI/UX wireframing), Canva
Automation & ops	Workflow automation, Visual Basic, SEO, digital strategy, Google Workspace, scheduling, technical documentation

Leadership & organizations

- **Partnerships & Linkages Director**, UP Computer Science Guild (UPCSG) · 2025 – 2026 — manages external partnerships, linkages, and sponsorships
- **Partnerships & Linkages Deputy**, UPCSG · 2024 – 2025
- **Representative**, Cebu Student Programmer's Association (CESPA) · 2025 – 2026
- **InterHigh Programming Competition Mentor**, UPCSG Komsai Week · 2025
- **Partnerships & Sponsorships Co-Head**, UPCSG Komsai Week · 2025
- **Technical Head**, UPCSG TekToks · 2023

Volunteer & community

Event Volunteer · AWS Student Community Day Cebu 2026 "Built to Build" — Aug 2026

- Handled event operations, stage logistics, and audience facilitation at the UP Cebu Performing Arts Hall for a regional gathering of student builders, speakers, and cloud practitioners.

Event Volunteer · AWS Community Day Cebu 2025, UP Cebu — Sep 2025

- Supported event operations and facilitation.

Event Volunteer & Organizer · QISTCon.ph 2025 — Jul 30 – Aug 1, 2025

- Supported organization, speakers, participants, and event flow at the Philippine Quantum Information, Science & Technology Conference, Jpark Island Resort, Cebu.

Trainings, seminars & certifications

- **Immersive Making: Art, Design, and Communication in VR/XR**, MANGO Workshop · Mar 9, 2026 · certificate
- **Panel on Research Ethics: Navigating Human Research**, Certificate Module · 2026
- **DevFest Cebu 2025**, GDG Cebu, Golden Peak Hotel and Suites · Nov 30, 2025 · certificate
- **AWS Mini Meetup**, University of the Philippines Cebu · Sep 14, 2025 · certificate
- **Cipher Cebu: Modern Cryptography Workshop 0**, UP Cebu · Sep 10, 2025
- **PBA-X: Introduction to Blockchain and Polkadot**, Polkadot Blockchain Academy (third wave) · Jul 25, 2025 · certificate
- **Notion Cebu PH: Lead Forward Year 2**, University of San Jose-Recoletos · May 18, 2025

References

- **Aliyah Khaet Regacho**, Virtual Assistant, Dilaab Digitals · 0966 223 1048 · liyahregacho@gmail.com
- **Algene T. Radaza**, SHS Research Coordinator, CIT-U · 0966 521 1020 · algene.radaza@cit.edu
- **Carl Lawrence Capon Leray**, Middle School Science Teacher, Jefferson Elementary School District · +1 832 837 5400 · cleray@jeffersonesd.org